

- For an integer $m > 1$, let $\mathcal{C}$ be the $[n, n - m, 3]$ Hamming code over $GF(2)$, where $n = 2^m - 1$. (The code is defined as having a parity check matrix which has all non-zero columns of length m .)
 - Show that every two distinct columns $\underline{h}_1$ and $\underline{h}_2$ in a parity-check matrix of $\mathcal{C}$, there is a unique third column in that matrix that equals the sum of $\underline{h}_1$ and $\underline{h}_2$.
 - Using the above part, show that the number of codewords of Hamming weight 3 in $\mathcal{C}$ is $\frac{n(n-1)}{6}$.
- Let H be the parity-check matrix of an (n, k) linear code $\mathcal{C}$. Let $\mathcal{C}'$ be the extended code whose parity-check matrix H' is formed by

$$H' = \begin{bmatrix} 0 & & & \\ \vdots & & H & \\ 0 & & & \\ 1 & \dots & & 1 \end{bmatrix}$$

- Show that every codeword in $\mathcal{C}'$ has even weight.
 - Let G be a generator matrix of $\mathcal{C}$. Specify a generator matrix of $\mathcal{C}'$.
- The generator matrix of a binary linear code $\mathcal{C}$ is given by

$$G = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- What is the minimum distance of $\mathcal{C}$?
 - Calculate the parity-check matrix of $\mathcal{C}$.
 - Suppose the channel output is the vector $\underline{y} = (1 \ 1 \ 1 \ 0)$, calculate the syndrome corresponding to this received vector. Also, calculate the coset to which $\underline{y}$ belongs.
 - What is the probability of error of the code under standard array decoding, if the channel is a binary symmetric channel with cross over probability p ?
- Let $\mathcal{C}$ be a linear $[n, k, d]$ code over $\mathbb{F}$ where $k > 1$. For $i \in \{1, 2, \dots, n\}$ denote by $\mathcal{C}^{(i)}$ the code

$$\mathcal{C}^{(i)} = \{(c_1, c_2, \dots, c_{i-1}, c_{i+1}, \dots, c_n) : (c_1, c_2, \dots, c_{i-1}, 0, c_{i+1}, \dots, c_n) \in \mathcal{C}\}.$$

The code $\mathcal{C}^{(i)}$ is said to be obtained by shortening $\mathcal{C}$ at the i^{th} coordinate.

- Show that $\mathcal{C}^{(i)}$ is a linear $[n - 1, k^{(i)}, d^{(i)}]$ code over $\mathbb{F}$ where $k^{(i)} \geq k - 1$ and $d^{(i)} \geq d$.
 - Show that a parity-check matrix of $\mathcal{C}^{(i)}$ is obtained by deleting the i^{th} column from a parity-check matrix of $\mathcal{C}$.
- Consider an $[n, k, d]$ linear code $\mathcal{C}$, over $\mathbb{F}_2$, with $d = 2t + 1$, for some $t \in \mathbb{N}$. Suppose that in the standard array corresponding to $\mathcal{C}$, there are no coset leaders of weight strictly larger than t . What, then, is the probability of error under standard array decoding, when $\mathcal{C}$ is used over a BSC with cross-over probability p ?